

Introduction to Abstract Algebra II: Homework 5

Due on February 17, 2026 at 23:59

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Exercise 31.2. Find all prime ideals and all maximal ideals of $\mathbb{Z}_{12}$.

Solution. The prime ideals of $\mathbb{Z}_{12}$ are given by $p_1^{k_1} \cdots p_n^{k_n}$, where each p_i is a prime factor of 12. Thus, for 12, we have $12 = 2^2 \cdot 3$. Thus, the prime ideals for $\mathbb{Z}_{12}$ are given by $\langle 2 \rangle$ and $\langle 3 \rangle$. Since we're working with $\mathbb{Z}_{12}$, the prime ideals are also maximal ideals. Thus, the maximal ideals of $\mathbb{Z}_{12}$ are also given by $\langle 2 \rangle$ and $\langle 3 \rangle$. $\square$

Exercise 31.4. Find all prime ideals and all maximal ideals of $\mathbb{Z}_2 \times \mathbb{Z}_4$.

Solution. The prime ideals of $\mathbb{Z}_2 \times \mathbb{Z}_4$ are given by $P \times \mathbb{Z}_4$ or $\mathbb{Z}_2 \times Q$, where P is a prime ideal of $\mathbb{Z}_2$ and Q is a prime ideal of $\mathbb{Z}_4$. The only prime ideal of $\mathbb{Z}_2$ is $\langle 0 \rangle$, and the prime ideals of $\mathbb{Z}_4$ is $\langle 2 \rangle$. Thus, the prime ideals of $\mathbb{Z}_2 \times \mathbb{Z}_4$ are given by $\langle 0 \rangle \times \mathbb{Z}_4$ and $\mathbb{Z}_2 \times \langle 2 \rangle$. The maximal ideals of $\mathbb{Z}_2 \times \mathbb{Z}_4$ are given by $M \times \mathbb{Z}_4$ or $\mathbb{Z}_2 \times N$, where M is a maximal ideal of $\mathbb{Z}_2$ and N is a maximal ideal of $\mathbb{Z}_4$. The only maximal ideal of $\mathbb{Z}_2$ is $\langle 0 \rangle$, and the maximal ideals of $\mathbb{Z}_4$ is $\langle 2 \rangle$. Thus, the maximal ideals of $\mathbb{Z}_2 \times \mathbb{Z}_4$ are given by $\langle 0 \rangle \times \mathbb{Z}_4$ and $\mathbb{Z}_2 \times \langle 2 \rangle$. $\square$

Exercise 31.6. Find all $c \in \mathbb{Z}_3$ such that $\mathbb{Z}_3[x]/\langle x^3 + x^2 + c \rangle$ is a field.

Solution. In order for $\mathbb{Z}_3[x]/\langle f_c(x) \rangle$ to be a field, where $f_c(x) = x^3 + x^2 + c$, we must have $f_c(x)$ be an irreducible polynomial. We check each of the values $c = 0, 1, 2$. For $c = 0$, we have $f_0(x) = x^3 + x^2$, which is reducible, since it has a root at $x = 0$. For $c = 1$, we have $f_1(x) = x^3 + x^2 + 1$, which is reducible, since it has a root at $x = 1$. For $c = 2$, we have $f_2(x) = x^3 + x^2 + 2$, which is irreducible, since it doesn't have any roots. Therefore, for $\mathbb{Z}_3[x]/\langle x^3 + x^2 + c \rangle$ to be a field, we must have $c = 2$. $\square$

Exercise 31.8. Find all $c \in \mathbb{Z}_5$ such that $\mathbb{Z}_5[x]/\langle x^2 + x + c \rangle$ is a field.

Solution. Again, just like problem 31.6, we check for what values of $c \in \mathbb{Z}_5$ give us $f_c(x) = x^2 + x + c$ an irreducible polynomial. Those values of c are 1 and 2, since $f_1(x)$ and $f_2(x)$ don't have any roots. $\square$

Exercise 31.16. Find a prime ideal of $\mathbb{Z} \times \mathbb{Z}$ that is not maximal.

Solution. We construct the ideal $I = p\mathbb{Z} \times \mathbb{Z}$, where $p = 2$ is a prime number. We now check that it is an ideal of $\mathbb{Z} \times \mathbb{Z}$.

Clearly, I is nonempty since $(2, 0) \in I$. Let $(a_1, b_1), (a_2, b_2) \in I$. Then $a_1, a_2 \in 2\mathbb{Z}$ and $b_1, b_2 \in \mathbb{Z}$. Hence

$$(a_1, b_1) - (a_2, b_2) = (a_1 - a_2, b_1 - b_2),$$

and since $2\mathbb{Z}$ and $\mathbb{Z}$ are ideals of $\mathbb{Z}$, we have $a_1 - a_2 \in 2\mathbb{Z}$ and $b_1 - b_2 \in \mathbb{Z}$. Thus I is closed under subtraction.

Now let $(r, s) \in \mathbb{Z} \times \mathbb{Z}$ and $(a, b) \in I$. Then $a \in 2\mathbb{Z}$, so $ra \in 2\mathbb{Z}$, and clearly $sb \in \mathbb{Z}$. Hence

$$(r, s)(a, b) = (ra, sb) \in I.$$

Thus I is an ideal of $\mathbb{Z} \times \mathbb{Z}$.

Next, we show that I is prime. Consider the quotient ring

$$(\mathbb{Z} \times \mathbb{Z})/I.$$

Define a map

$$\varphi : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}, \quad \varphi(a, b) = \bar{a}.$$

This is a surjective ring homomorphism, and

$$\ker \varphi = 2\mathbb{Z} \times \mathbb{Z} = I.$$

By the First Isomorphism Theorem,

$$(\mathbb{Z} \times \mathbb{Z})/I \cong \mathbb{Z}/2\mathbb{Z},$$

which is a field, hence an integral domain. Therefore I is a prime ideal.

Finally, I is not maximal. Indeed, consider the ideal

$$J = 2\mathbb{Z} \times 2\mathbb{Z}.$$

Then

$$I = 2\mathbb{Z} \times \mathbb{Z} \subsetneq 2\mathbb{Z} \times 2\mathbb{Z} \subsetneq \mathbb{Z} \times \mathbb{Z},$$

so there exists a proper ideal strictly between I and the whole ring. Hence I is not maximal.

Therefore $I = 2\mathbb{Z} \times \mathbb{Z}$ is a prime ideal of $\mathbb{Z} \times \mathbb{Z}$ that is not maximal. $\square$

Exercise 31.18. Is $\mathbb{Q}[x]/\langle x^2 - 5x + 6 \rangle$ is a field? Why?

Solution. The quotient $\mathbb{Q}[x]/\langle x^2 - 5x + 6 \rangle$ isn't a field since we can factor the polynomial into $(x-2)(x-3)$. Since the polynomial is reducible, the ideal $\langle x^2 - 5x + 6 \rangle$ isn't maximal, meaning the quotient isn't a field. $\square$

Exercise 31.24. Give an example of an ideal in $\mathbb{Q}[x, y]$ that is not a principal ideal. Conclude that if R is an integral domain with the property that every ideal in R is principal, it does not follow that every ideal in $R[x]$ is a principal ideal.

Solution. Consider the ideal

$$I = \langle x, y \rangle \subset \mathbb{Q}[x, y].$$

We claim that I is not principal. We prove this via contradiction. Assume that $I = \langle f(x, y) \rangle$ for some $f \in \mathbb{Q}[x, y]$. Since $x, y \in I$, we must have

$$f \mid x \quad \text{and} \quad f \mid y$$

Thus f is a common divisor of x and y . However, x and y are irreducible and not associates in the UFD $\mathbb{Q}[x, y]$, so their only common divisors are units. Hence f must be a unit, which implies

$$I = \mathbb{Q}[x, y],$$

a contradiction since $1 \notin \langle x, y \rangle$. Therefore $\langle x, y \rangle$ is not a principal ideal.

Now let R be an integral domain such that every ideal in R is principal. Taking $R = \mathbb{Q}$, which is a field (hence a PID), we have

$$R[x][y] = \mathbb{Q}[x, y].$$

Since $\mathbb{Q}[x, y]$ contains the non-principal ideal $\langle x, y \rangle$, it follows that even if every ideal in R is principal, it does not follow that every ideal in $R[x]$ is principal. $\square$

Exercise 31.26. Let R be a finite commutative ring with unity. Show that every prime ideal in R is a maximal ideal.

Solution. Let P be a prime ideal in R . Then the quotient ring R/P is an integral domain. Since R is finite and P is an ideal, the quotient R/P is also finite. Thus R/P is a finite integral domain. We claim that every finite integral domain is a field. Let $\bar{a} \in R/P$ with $\bar{a} \neq 0$. Consider the map

$$\varphi : R/P \rightarrow R/P, \quad \varphi(\bar{x}) = \bar{a}\bar{x}.$$

If $\varphi(\bar{x}) = \varphi(\bar{y})$, then

$$\bar{a}\bar{x} = \bar{a}\bar{y} \Rightarrow \bar{a}(\bar{x} - \bar{y}) = 0.$$

Since R/P is an integral domain and $\bar{a} \neq 0$, it follows that $\bar{x} = \bar{y}$. Thus φ is injective. Because R/P is finite, φ is also surjective. Hence there exists $\bar{x} \in R/P$ such that

$$\bar{a}\bar{x} = 1$$

Therefore every nonzero element of R/P is invertible, so R/P is a field.

Since R/P is a field, P is a maximal ideal. Thus every prime ideal in R is maximal. $\square$

Exercise 31.32. Prove that if F is a field, every proper nontrivial prime ideal of $F[x]$ is maximal.

Solution. Let P be a proper nontrivial prime ideal of $F[x]$, where F is a field. Since $F[x]$ is a principal ideal domain, there exists a nonzero polynomial $f(x) \in F[x]$ such that

$$P = \langle f(x) \rangle.$$

Because P is prime, $\langle f(x) \rangle$ is a prime ideal. In a PID, a principal ideal is prime if and only if its generator is irreducible. Thus $f(x)$ is irreducible in $F[x]$. Consider the quotient ring

$$F[x]/\langle f(x) \rangle.$$

Since $f(x)$ is irreducible over the field F , the ideal $\langle f(x) \rangle$ is maximal, and hence the quotient is a field.

Therefore P is a maximal ideal. Thus every proper nontrivial prime ideal of $F[x]$ is maximal. $\square$

Exercise 31.36. If A and B are ideals of a ring R , the *sum* $A + B$ of A and B is defined by

$$A + B = \{a + b \mid a \in A, b \in B\}.$$

(i) Show that $A + B$ is an ideal.

(ii) Show that $A \subseteq A + B$ and $B \subseteq A + B$.

Solution to (i). First, $0 = 0 + 0 \in A + B$ since $0 \in A$ and $0 \in B$. Let $x, y \in A + B$. Then there exist $a_1, a_2 \in A$ and $b_1, b_2 \in B$ such that

$$x = a_1 + b_1, \quad y = a_2 + b_2.$$

Then

$$x - y = (a_1 - a_2) + (b_1 - b_2).$$

Since A and B are ideals, $a_1 - a_2 \in A$ and $b_1 - b_2 \in B$. Hence $x - y \in A + B$. Thus $A + B$ is closed under subtraction.

Now let $r \in R$ and $x = a + b \in A + B$. Then

$$rx = ra + rb.$$

Since A and B are ideals, $ra \in A$ and $rb \in B$, so $rx \in A + B$. Similarly, $xr \in A + B$ if R is not assumed commutative.

Therefore $A + B$ is an ideal of R . $\square$

Solution to (ii). Since for any $a \in A$ we have $a = a + 0$ with $0 \in B$, it follows that $a \in A + B$. Thus $A \subseteq A + B$.

Similarly, for any $b \in B$ we have $b = 0 + b$ with $0 \in A$, so $b \in A + B$. Hence $B \subseteq A + B$. $\square$